

DESWIK.SCHED FULL TRAINING REPORT

From First Principles to Underground Medium-Term Planning

General Underground Scheduling + 25-Stope MTP Model

Deswik.Sched 2026.1 | July-August 2026

25 Stopes	1.986 Mt Planned ore	6 Final DEV streams	24 Jul 2027 Final completion
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Prepared for portfolio, competency evidence, and technical reflection

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Document Purpose and Control

This report documents the complete Deswik.Sched learning journey from the first general underground scheduling exercise through the final resource-constrained medium-term planning model. It is both a training record and an evidence-based explanation of the planning concepts, troubleshooting methods, reports, assumptions, and conclusions developed during the project.

Item	Detail
Software	Deswik.Sched
Training model	Synthetic underground longhole stoping schedule
Formal MTP horizon	1 August 2026 to 31 July 2027
Final model size	25 stopes, 3 panels, 4 horizons, 5 sublevels
Final baseline	24 July 2027, 25 stopes, 1,986,000 t
Updated forecast	24 July 2027 after progress update and re-leveling
Important limitation	Synthetic data and unverified geotechnical sequencing; not suitable for operational issue or construction

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EXECUTIVE SUMMARY

Executive Summary

The training began with a simple underground schedule designed to learn Deswik.Sched fundamentals: task hierarchy, predecessors, calendars, resources, leveling, reporting, progress entry, and SViz. That first model was intentionally limited. It revealed that increasing production resources did not materially improve completion because production tasks were not sufficiently overlapped or ready. The controlling issue was development exposure, not the production fleet.

The learning then moved into a purpose-built medium-term planning model with 25 stopes across North, Central, and South panels, four vertical horizons, and five sublevels. The final model used native production quantities, resource assignment rate tables, explicit priorities, monthly production reports, readiness inventories, and a progress-update forecast.

Final validated result

The selected baseline uses three additional development heading streams, increasing the original pool from 3 to 6. It delivers all 25 stopes and 1.986 Mt by 24 July 2027. The updated forecast also completes on 24 July 2027, showing that intermediate milestone movement was absorbed after correct re-leveling.

Major learning	Evidence
Primary resource constraint	DEV_HEADING_STREAM; increasing development capacity improved completion until diminishing returns.
Optimal capacity	+3 development streams selected as the practical point; +5 was a plateau test, not the final recommendation.
Major data-quality defect	Four stope packages had Priority = 0, causing artificial starvation and a three-month production gap.
Readiness interpretation	Month-end Development Ready or similar inventory can be normal lead time, not a queue, if it converts in the next month.
Event reporting	Use native Task Count (Finish In) and Task Count (Start In), not a progressive custom event quantity.
Forecast discipline	Always update actual status accurately and re-level before judging completion, resource queues, or inventory.

PART I - GENERAL TRAINING

1. Training Starting Point

1.1 Initial questions

The training started from practical questions rather than a predefined course. These questions shaped the entire learning path:

- What does a one-week planning horizon actually mean in a real schedule?
- Is Deswik.Sched only useful for short-term planning, or can it support MTP?
- How do WBS parents, detailed tasks, dependencies, calendars, and resources interact?
- Why does increasing a production resource sometimes produce no schedule improvement?
- How can a report distinguish a real bottleneck from normal process lead time?
- What can SViz do without Deswik.CAD integration?

1.2 Planning-horizon understanding

Horizon	Purpose	Typical schedule character
Long-term planning	Strategic value, mine shape, reserves, major phases and capital direction	Coarse periods, strategic constraints, optimization and design integration
Medium-term planning	Translate strategy into executable development, stoping, fleet and monthly production plans	Resource-constrained tasks, monthly quantities, readiness and replacement inventory
Short-term planning	Detailed execution for weeks, shifts, headings and crews	High detail, frequent updates, close plan-versus-actual control

Key correction

A one-week horizon does not mean the entire mine schedule is recreated from zero every week. It means the near-term plan is refreshed at a finer level of detail while the broader schedule remains the reference.

1.3 What Deswik.Sched is good at

The training corrected the early assumption that Deswik.Sched is only a short-term tool. It is strongest wherever detailed, resource-constrained activities must be sequenced and updated. That includes short-term execution and many medium-term planning applications. It is not, by itself, a strategic pit or stope optimizer and does not replace mine design, geotechnical validation, or economic optimization.

2. General Deswik.Sched Training

2.1 WBS and task hierarchy

The first technical skill was structuring a mine plan into a hierarchy that separates reporting parents from executable activities. A typical chain was built around primary access, level development, ventilation and services, footwall drives, crosscuts, drill drives, stope preparation, production drilling, blasting, bogging, backfill, curing, and completion milestones.

Layer	Example	Purpose
Project / panel parent	L1200 Development, North Panel	Roll-up, filtering and reporting
Location package	N-H1-01 stope package	Connects access and production work for one stope
Executable activity	Top crosscut, production drilling, bogging	Receives duration, quantity, resource, calendar and predecessors
Milestone	Development Ready, Drill Ready, Stope Complete	Marks a readiness gate without consuming a resource

2.2 Dependency logic

The training used Finish-Start relationships for most physical sequence logic and WBS Parent links for hierarchy. The most important lesson was that a high priority cannot override an incomplete predecessor, a non-working calendar, or unavailable resource. A task must first be eligible before priority can decide which task wins.

Dependency discipline

A schedule can look resource-constrained when the real cause is a predecessor. Physical, geotechnical, access, and administrative dependencies must be separated and verified.

2.3 Calendars, duration and active versus elapsed work

Active work and elapsed waiting were separated. Development, drilling, bogging, services, and paste placement use working calendars and resources. Passive curing uses elapsed time and no resource. This distinction prevented cure time from incorrectly consuming paste-fill capacity.

Type	Examples	Modeling treatment
Active production	Development, drilling, bogging, paste placement	Production quantity + assignment rate + working calendar + resource
Active time-based work	Ventilation, services, cablebolt preparation, charge/blast/re-entry	ACT_D quantity at 1 activity-day/day + resource
Elapsed waiting	Cure	Fixed elapsed duration, no resource
Event	Development Ready, Drill Ready, Complete	Zero-duration milestone

2.4 Resources, rates, and the first major troubleshooting lesson

A crucial discovery was that, in this configuration, the rate assigned with the resource or resource rate table controlled the effective task duration. Entering a production quantity without the correct assignment rate created unrealistic schedules, including a plan extending to the year 2170. The correction was to make the resource assignment rate table the source of truth and verify every rate after assignment.

Problem observed	Root cause	Correction
Resource assignment produced zero duration	Task had zero driving quantity	Add a native quantity such as ACT_D for active time-based work
Plan extended for centuries	Assignment rate defaulted to an incorrect low value	Link each resource assignment to the correct rate table
Task rate looked correct but schedule was wrong	Visible task-rate field was not controlling the assigned resource	Validate assignment-level effective rate and resulting duration

2.5 Resource leveling and priority

The general training introduced the Resource Leveling window, scheduling priority rules, and the difference between task eligibility and task ranking. Priority was configured as a numeric field in descending order, so higher numbers win among eligible tasks competing for the same pool.

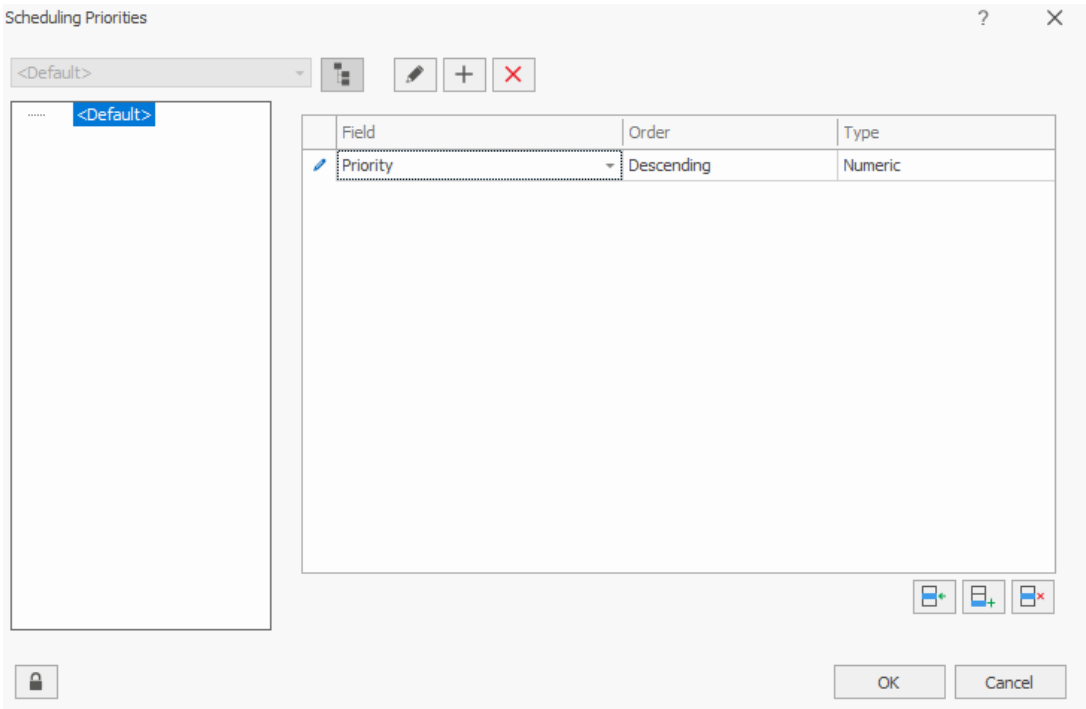


Figure 1. Scheduling priority rule configured as numeric and descending.

Rule	Meaning
Higher priority number	Wins only when tasks are simultaneously eligible and compete for the same constrained resource
Clear leveling before a test	Removes the previous allocation result so the scenario is comparable
One-variable sensitivity	Change only one resource or rule at a time
No task splitting in baseline	Keeps the first baseline easier to interpret

2.6 General-training bottleneck result

The first schedule did not provide a strong production bottleneck test because the production tasks were mostly serial and were not physically ready at the same time. Increasing a production drill, LHD, or other production resource separately produced little improvement. Development exposure was the real constraint: upper stopes could progress while lower blocks still waited for development, so added production equipment had no simultaneous work to process.

Why the general training succeeded

The model was not a failed bottleneck study. It proved the ability to inspect the schedule independently, test resource quantities, reject unsupported conclusions, and identify that the training horizon and task overlap were insufficient for a full production bottleneck analysis.

2.7 Reports, formulas, and data date

Topic	Learning
Report formulas	SGN returns direction; MAX can prevent negative outputs; ABS removes direction. For 0/1 task-count flags, threshold logic must reflect the actual value range.
Progress date	Use a data date and update actual status; do not complete tasks only because their planned date is earlier.
Re-level from date	Preserve historical results and level only unfinished future work.
Report portability	Report options can be reused, but fields, filters and calendars must exist consistently in the destination project.

2.8 General-training outcome

By the end of the general training, the key transition was from using Deswik.Sched as a date calculator to using it as a logical model. The schedule was understood as an interaction between physical readiness, precedence, calendars, resource capacity, rate assumptions, priorities, and reporting definitions.

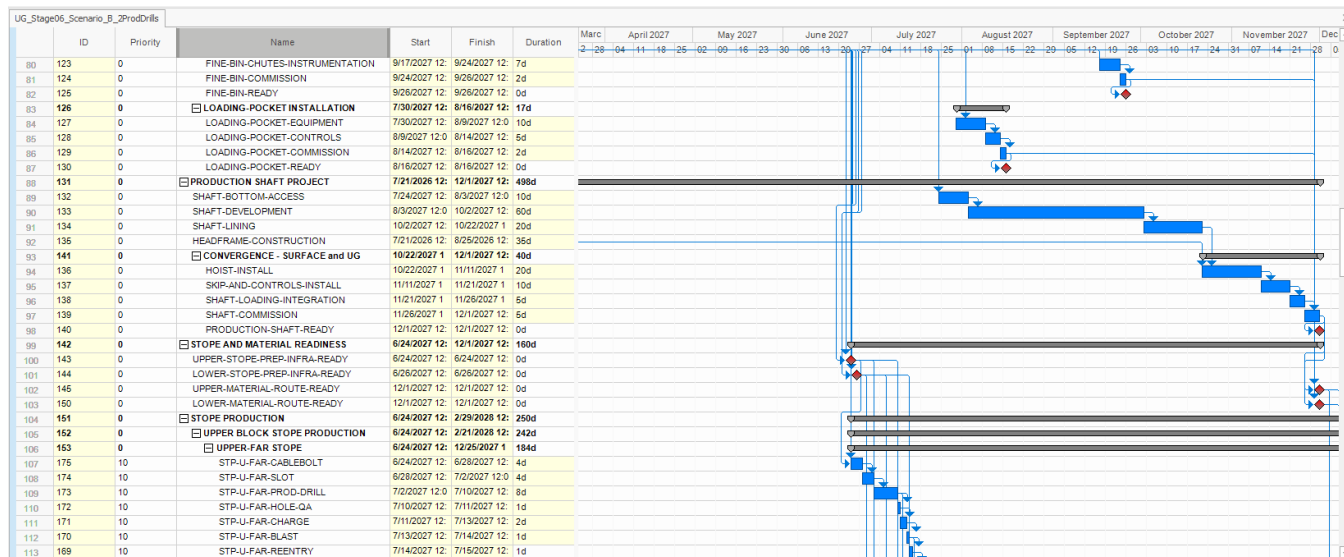


Figure 2. General training task grid view

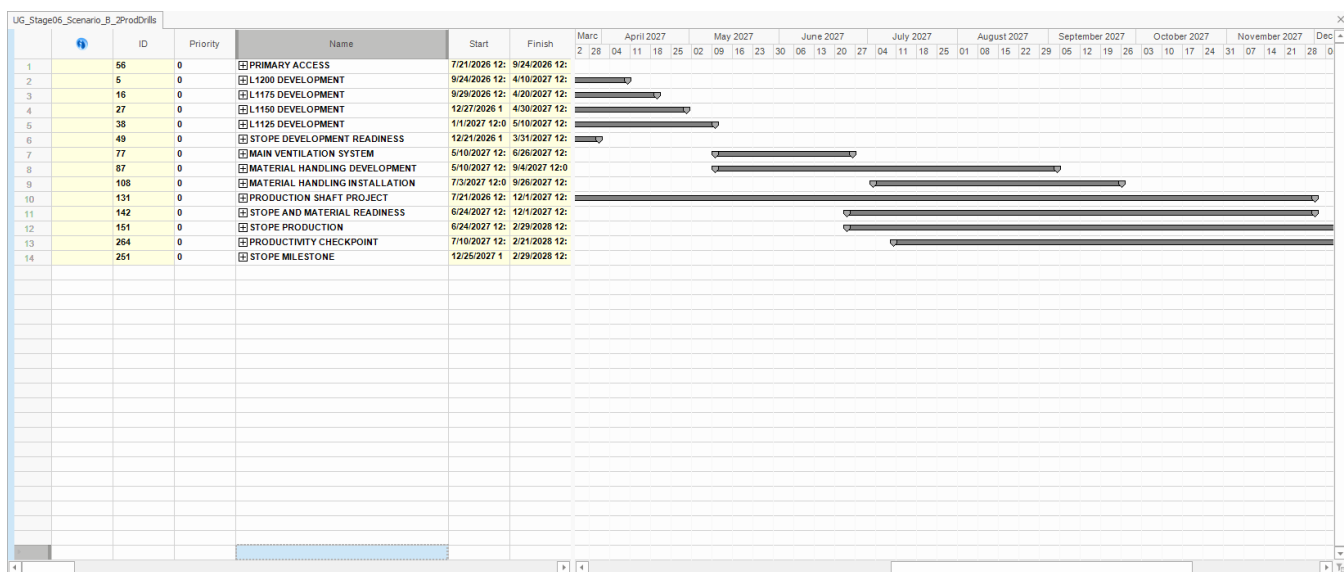


Figure 3. General training parent task

PART II - UNDERGROUND MEDIUM-TERM PLANNING

3. Transition to a Purpose-Built MTP Model

The MTP model was created because the first training schedule did not contain enough simultaneous headings and stopes to test tactical production behavior properly. The new model deliberately introduced multiple panels, horizons, resource pools, and readiness gates so that development, preparation, production, fill, and sequencing could compete over a 12-month planning horizon.

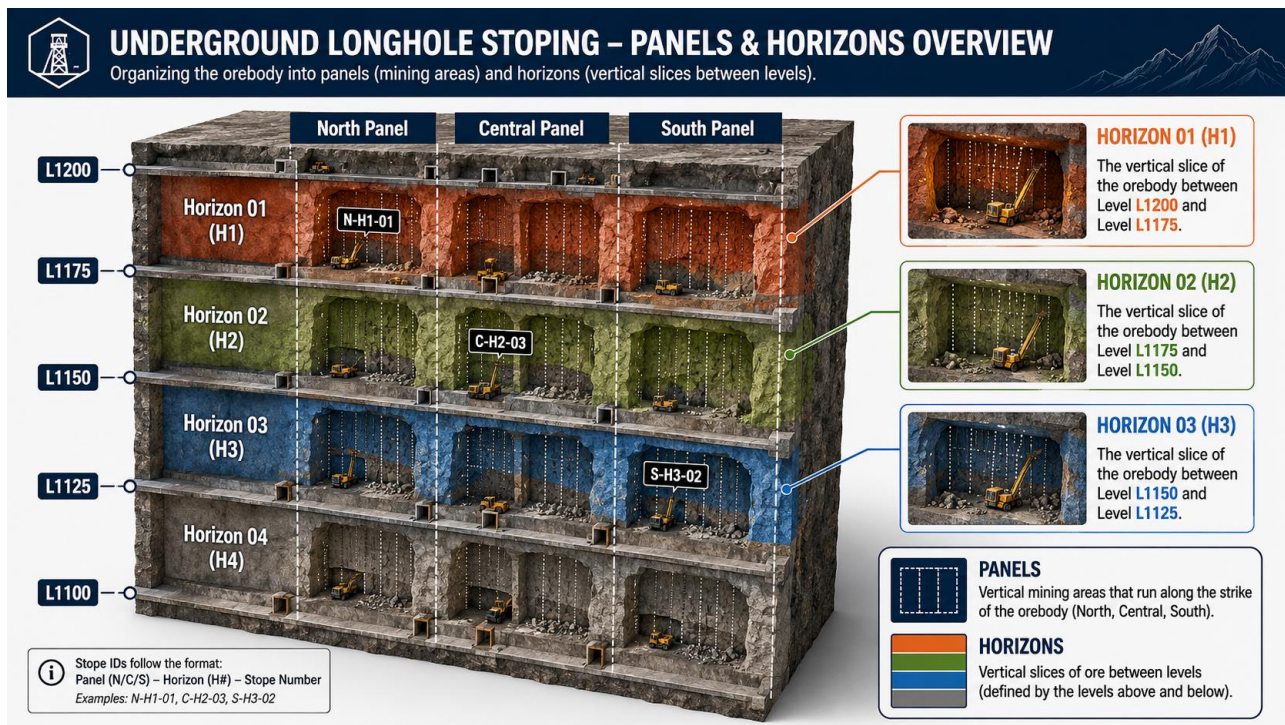


Figure 4. Synthetic layout concept: three panels across four vertical production horizons.

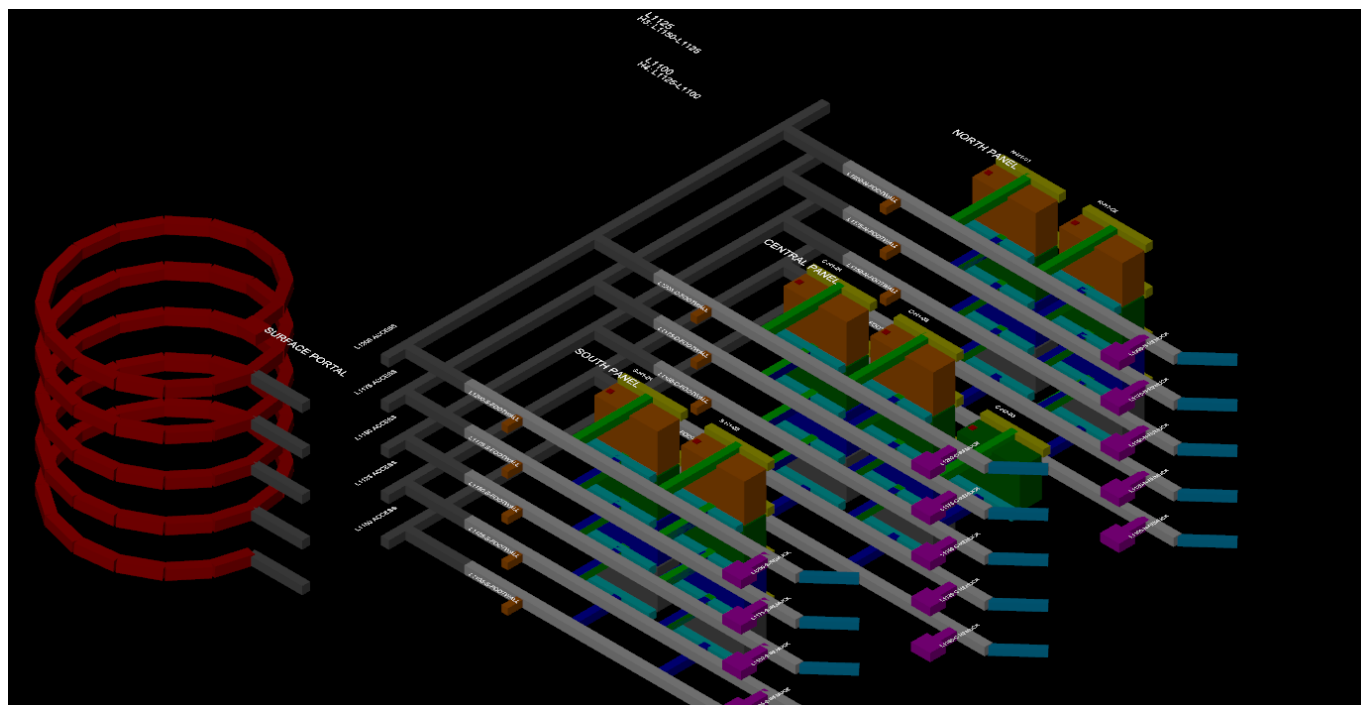


Figure 5. Synthetic underground design: three panels across four vertical production horizons and five sublevels.

3.1 Mine and schedule architecture

Dimension	Final model
Sublevels	L1200, L1175, L1150, L1125, L1100
Horizons	H1 to H4, each between two adjacent sublevels
Panels	North, Central and South
Stopes	25 total: North 8, Central 9, South 8
Naming convention	Panel-Horizon-Stope, for example N-H1-01
Formal period	August 2026 to July 2027

Intermediate levels have two functions. For example, L1175 is the lower extraction level for H1 and the upper drilling level for H2. The same elevation supports different headings and work packages.

HOW ONE LEVEL CAN ACT AS BOTH

Intermediate sublevels can be the extraction level for the horizon above and the drill level for the horizon below.

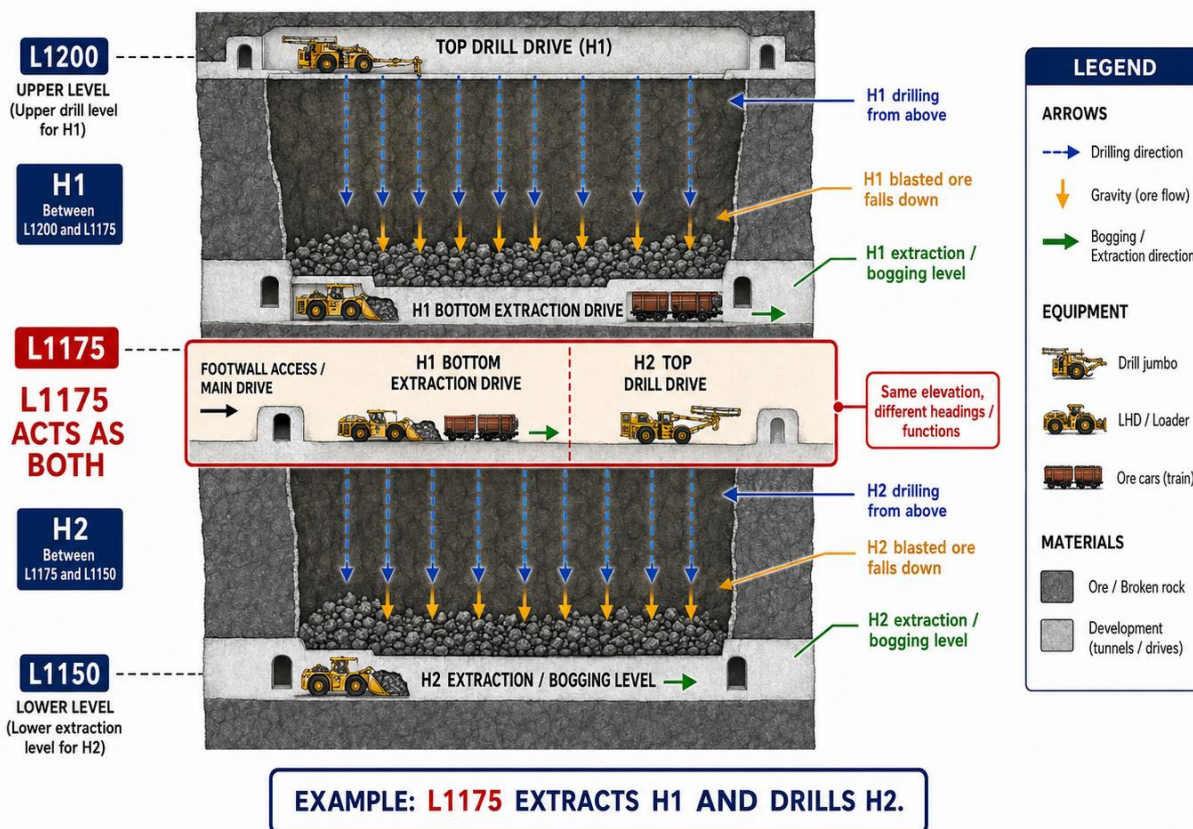


Figure 6. Intermediate sublevel serving the lower horizon and the upper horizon at the same elevation supported by sill pillar in between.

3.2 Chronological stage map

Stage	Focus	What was learned
0-1	Import foundation and integrity	Import Name, Duration, Predecessor Details and Description; validate hierarchy and logic.
2	Classification and coding	Create workstream, activity class, panel, level/horizon and location fields.
3	Calendars	Separate active work calendars from elapsed curing and add shutdown assumptions.
4	Native production and rates	Use DEV_M, SLOT_M, DRILL_M, ORE_T, FILL_M3 and ACT_D.
5	Resources	Assign composite work streams and rate tables; split paste placement from cure.
6	Priorities and leveling	Create production-first sequence and validate descending priority behavior.
7	Monthly reporting	Time-phase physical quantities and compare baseline scenarios.
8	Readiness inventory	Use Finish In and Start In task counts to

Stage	Focus	What was learned
9	Bottleneck validation	calculate state inventory. Test resource quantities one at a time; distinguish lead time from queue.
10	Capacity selection	Choose +3 DEV streams as the practical optimum.
11	Baseline freeze	Create the formal monthly production and readiness plan.
12	Progress update	Enter actual progress, level from the data date, and compare forecast.
13	Recovery framework	Defined as an optional controlled-delay test; not executed as a separate final scenario.
14	Audit and handover	Document KPIs, assumptions, risks, management summary and final files.

4. Import, Classification, and Native Fields

4.1 Import limitation and workflow

The exported Deswik CSV exposed many read-only properties. Only four fields were reliably editable for import: Name, Duration, Predecessor Details, and Description. Therefore, the import created the task network and all other schedule configuration was completed inside Deswik.Sched.

Practical import lesson

Do not design a workflow around writing values into exported read-only fields. Import the minimum reliable network, then configure production fields, calendars, resources, priorities, and reporting natively.

4.2 Classification fields

Field	Purpose	Examples
WORKSTREAM	High-level process filter	DEVELOPMENT, STOPE_PREP, PROD_DRILL, BOGGING, BACKFILL
ACTIVITY_CLASS	Detailed activity type	FOOTWALL_DRIVE, SLOT_RAISE, PRODUCTION_DRILL, CURE
PLAN_AREA	Panel or shared area	NORTH, CENTRAL, SOUTH, SHARED
LEVEL_HORIZON	Vertical planning area	L1200-L1100 or H1-H4
LOCATION_ID	Exact package identifier	N-H1-01

4.3 Native production fields

Field	Abbreviation	Use	Full-model total
Development Metres	DEV_M	Horizontal access and stope development	6,225 m
Slot Metres	SLOT_M	Slot raises	625 m
Production Drill Metres	DRILL_M	Longhole production drilling	70,929 m
Stope Ore Tonnes	ORE_T	Progressive bogging production	1,986,000 t
Paste Fill Volume	FILL_M3	Active paste placement	674,491 m3
Activity Work Days	ACT_D	Active time-based tasks at 1 day/day	Task-specific

The Driving Property selected the appropriate production abbreviation. Driving Quantity was read-only and was populated from the native field. Duration then followed the assigned resource rate.

5. Resource Model and Effective Rates

Resource pool	Initial quantity	Role
DEV_HEADING_STREAM	3	Composite development cycle stream; final selected quantity 6
VENT_CREW	1	Ventilation establishment and extensions
SERVICES_CREW	2	Services establishment and extensions
STOPE_PREP_CREW	2	Cablebolt and preparation work
SLOT_RIG	1	Slot raise development
PROD_DRILL_FLEET	3	Production drilling
CHARGE_CREW	2	Charge, blast and re-entry
PROD_LHD_STREAM	3	Bogging and CMS work
PASTE_FILL_STREAM	2	Active paste fill placement

The development stream was intentionally modeled as a composite tactical resource rather than a literal jumbo count. Its effective development rate represented the full cycle needed to advance a heading.

5.1 Backfill correction

The original combined barricade-paste-cure activity incorrectly mixed active paste placement with passive cure time. It was split into: Bogging -> Paste Fill -> Cure -> Stope Complete. Paste Fill used FILL_M3 and a paste resource; Cure remained a fixed elapsed task without a resource.

6. Priority Strategy and Schedule Integrity

6.1 Why production-first leveling changed the schedule

Default leveling initially spread three development streams across L1200 North, Central and South. The production-first strategy redirected one stream to L1175 North so the upper and lower access required for the first H1 stope could be completed together. This advanced the first production-ready stope rather than maximizing broad but incomplete development exposure.

Milestone	Default	Production-first	Change
First Development Ready	12 Nov 2026	9 Aug 2026	About 3 months earlier
First Production Drilling	19 Dec 2026	16 Sep 2026	About 3 months earlier
First Blast	30 Dec 2026	25 Sep 2026	About 3 months earlier
First Bogging	1 Feb 2027	28 Sep 2026	About 4 months earlier
First H2 Blast	28 Mar 2027	31 Jan 2027	About 2 months earlier
First H3 Blast	5 Jul 2027	20 May 2027	About 1.5 months earlier

6.2 The priority-data defect

A three-month production interruption from August to October 2027 was traced through the reports. Production drilling continued, but blast count, bogged tonnes, filled stopes, and completed stopes did not increase. Inspection found Priority = 0 on four critical packages: C-H2-03, N-H3-02, C-H3-02, and S-H3-02.

Evidence	Interpretation
No blast, bogging, fill or completion increments	The chain was blocked at blast release, not at drilling productivity.
C-H2-03 delayed to October	Its cure predecessor delayed C-H3-01 and the following sequence.
All H3-02 packages also had Priority = 0	The issue affected the broader tail, not only the Central panel.
Correcting priorities improved completion	Confirmed a data-quality/configuration defect rather than a physical equipment shortage.

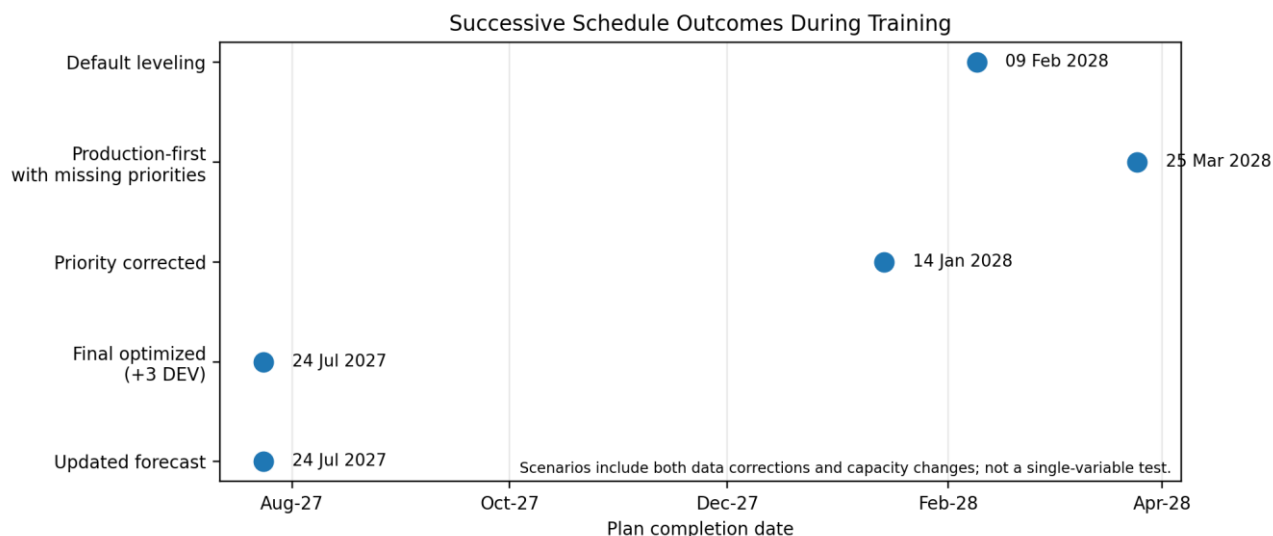


Figure 7. Successive completion outcomes as schedule logic, priority data, and development capacity were improved.

Geotechnical caution

The C-H2-03 Cure -> C-H3-01 Blast dependency was retained as an unverified geotechnical constraint. A schedule benefit is not sufficient reason to remove a curing or exposure dependency.

7. Monthly Production Reporting

Monthly reports converted the schedule into an MTP production profile. Physical quantities were time-phased over active task duration. Parent and summary tasks were excluded to prevent double counting. Full-schedule totals validated the reporting setup.

Output	Reporting basis	Validation total
Development metres	Progressive DEV_M on lowest-level development tasks	6,225 m
Slot metres	Progressive SLOT_M on slot raise tasks	625 m
Production drilling	Progressive DRILL_M	70,929 m
Bogged ore	Progressive ORE_T on bogging tasks	1,986,000 t
Paste fill	Progressive FILL_M3 on active paste tasks	674,491 m3
Blast / complete count	Task Count (Finish In) with activity filters	25 each

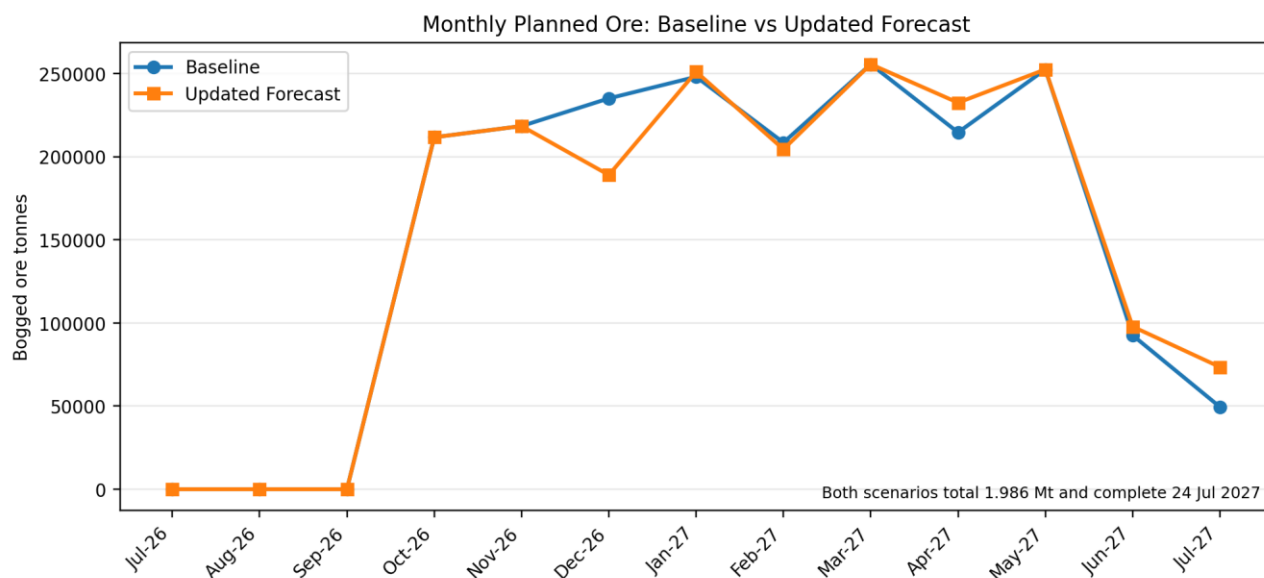


Figure 8. Baseline and updated monthly bogged-ore profiles. Timing changes, but both deliver 1.986 Mt.

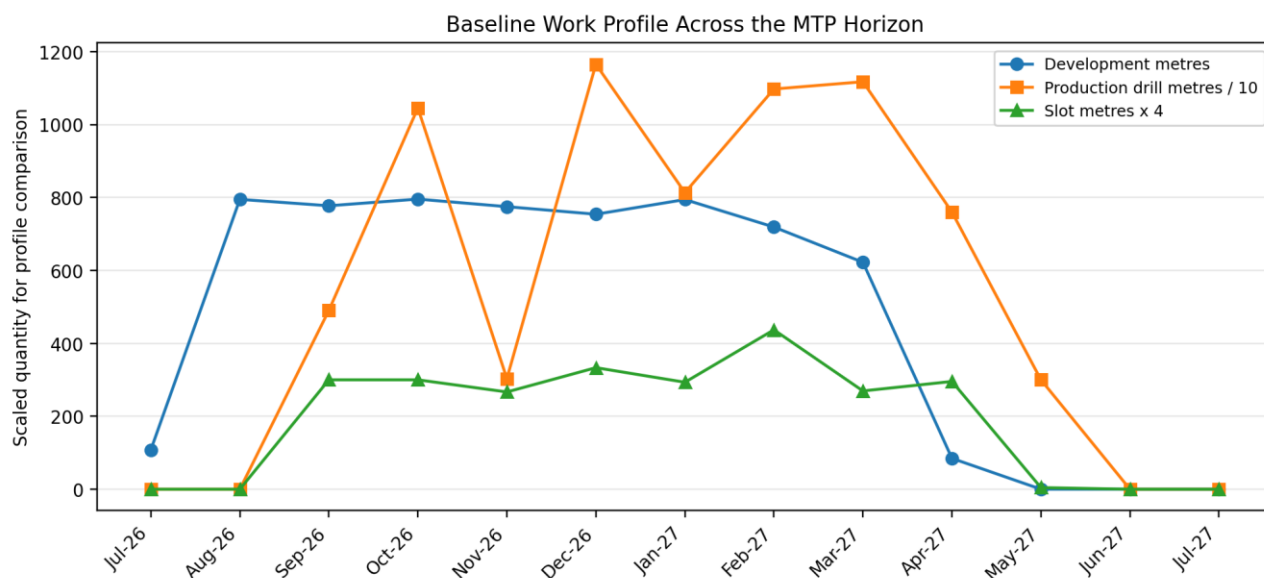


Figure 9. Baseline development, slot, and production-drilling profiles across the planning horizon.

8. Readiness Inventory: Finish In, Start In, and Lead Time

8.1 Native event counting

A custom Stope Event Count field did not work because its default time-cycle behavior progressively distributed the quantity. The correct method was to use native Task Count (Finish In) and Task Count (Start In) with activity-class filters. Each inventory required two cumulative report IDs and one final formula ID.

Combination	Meaning	Example
Finish - Start	Waiting only	Drill Ready Finish - Production Drilling Start = stopes waiting for the drill
Finish - Finish	Waiting plus work in progress	Drill Ready Finish - Production Drilling Finish
Start - Finish	Work in progress only	Production Drilling Start - Production Drilling Finish

8.2 Core inventory formulas

Inventory	Formula
Development-ready only	Cumulative Development Ready Finish - Cumulative Drill Ready Finish
Waiting for production drill	Cumulative Drill Ready Finish - Cumulative Production Drilling Start
Currently drilling	Cumulative Production Drilling Start - Cumulative Production Drilling Finish
Drilled, waiting for charge	Cumulative Production Drilling Finish - Cumulative Charge/Blast Start
Waiting for fill	Cumulative Boggging Finish - Cumulative Paste Fill Start
Curing	Cumulative Paste Fill Finish - Cumulative Stope Complete Finish
Broken ore remaining	Cumulative blasted tonnes - cumulative progressive bogged tonnes

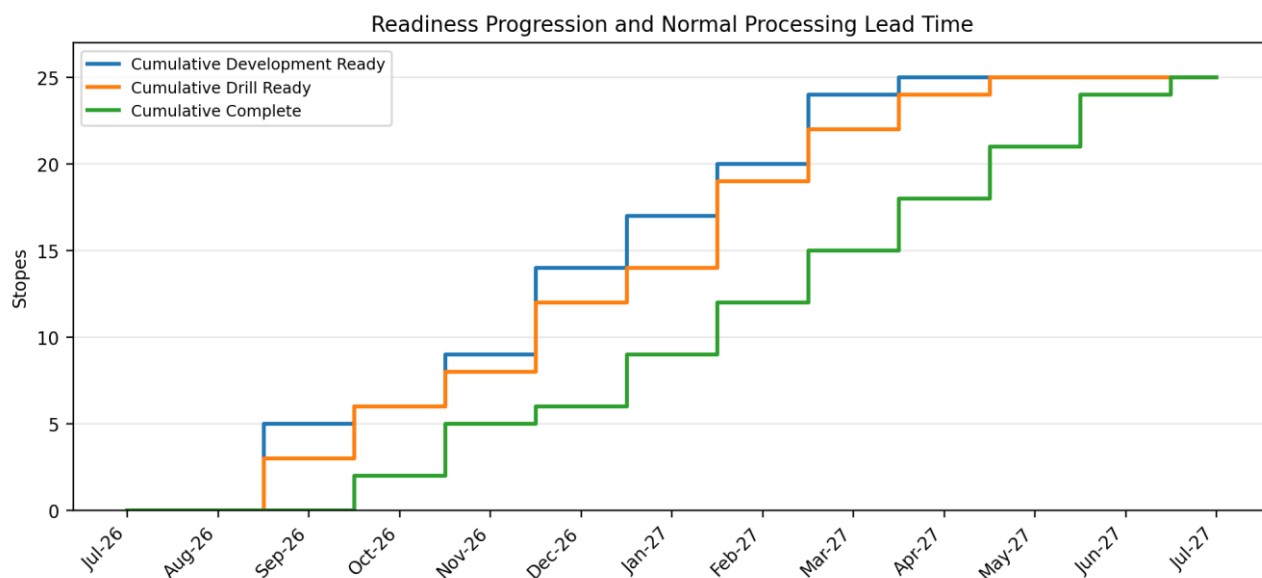


Figure 10. Baseline cumulative readiness gates. The temporary gap between gates reflects processing lead time.

8.3 Lead time is not automatically a bottleneck

Development Ready Only inventory appeared at some month ends because stopes became development-ready in the middle or late part of the month. Slot raising and cablebolt preparation then completed in the following month. The next report showed Development Ready Only decreasing and Drill Ready increasing. This was normal lead time, not evidence that the slot rig or stope preparation crew was constrained.

Diagnostic rule

A real queue persists or accumulates over multiple periods while the downstream gate grows slowly. A one-period inventory that converts in the following month is usually process lead time.

8.4 Coverage months

Prepared-production coverage was calculated using the current prepared tonnes divided by the average planned bogged tonnes of the next three months. For example, May prepared tonnes were compared with the average of June, July, and August

bogging. Three months was used to smooth the lumpy stope production profile; it was a training assumption, not a universal rule.

9. Bottleneck Analysis and Capacity Selection

9.1 Confirmed primary constraint

One-variable resource tests confirmed DEV_HEADING_STREAM as the primary resource constraint. Additional development capacity improved plan completion, while production resources produced little or no material improvement when physical readiness or predecessor constraints remained.

9.2 Constraint migration and plateau

The development pool was increased in steps. More development streams improved the schedule until the benefit flattened. Adding up to five extra streams was used to identify the plateau; however, the final baseline selected +3 streams because this was the practical optimal point. The final pool therefore increased from 3 to 6.

Observation	Conclusion
DEV +3 captured the practical benefit	Selected final capacity: 6 total streams
Further additions produced diminishing or no completion gain	Development was no longer the controlling limit
Slot and stope-prep inventories converted in the following month	No persistent downstream preparation constraint
Extra charge crew did not improve the correctly re-levelled finish	Charge crew not confirmed as the primary constraint

9.3 Bottleneck definition learned

Bottleneck definition

The bottleneck is not simply the busiest resource. It is the constraint whose additional capacity or corrected logic materially improves the selected objective: tonnes, readiness, milestone date, production continuity, or final completion.

10. Progress Update and Forecasting

The final training simulated a progress update to 31 October 2026. The baseline remained protected, actual status was entered, and the remaining work was re-levelled from 1 November 2026.

KPI	Baseline	Updated Forecast	Variance
First H2 blast	3 Dec 2026	7 Dec 2026	+4 days
First H3 blast	2 Feb 2027	6 Feb 2027	+4 days
Ore by July 2027	1,986,000 t	1,986,000 t	0
Completed stopes by July 2027	25	25	0
Full-plan completion	24 Jul 2027	24 Jul 2027	0 days

An apparent 27 July completion and excessive waiting-for-blast inventory were initially observed because the schedule had not been re-levelled after the latest changes. Once correctly re-levelled with the original charge crew quantity, the final completion returned to 24 July and the waiting inventory reduced.

Forecasting control lesson

Never judge schedule completion or inventory immediately after changing progress, priorities, rates, or resource quantities. Re-level first, then compare against the frozen baseline.

The four-day movement in intermediate blast milestones was absorbed later in the schedule. This demonstrated that a milestone variance does not automatically become a final completion variance.

11. Final Baseline and Management Handover

Final KPI	Result
Stopes	25
Planned ore	1,986,000 t
Development metres	6,225 m
Slot metres	625 m
Production drill metres	70,929 m
Paste-fill volume	674,491 m ³
Selected development capacity	6 streams (original 3 + 3)
Baseline completion	24 July 2027
Updated forecast completion	24 July 2027
Final completion variance	0 days

11.1 Management-ready conclusion

The final underground MTP schedules 25 stopes across three panels and four production horizons, delivering 1.986 Mt by 24 July 2027. Development heading capacity was confirmed as the primary resource constraint. Three additional development streams were selected as the practical optimum. Slot raising, stope preparation, charging, bogging, and paste filling were not shown to govern final completion under the validated schedule.

The most important disruption in the training model was not equipment capacity but missing priority data on four stope packages. Correcting the priorities restored blasting, bogging, filling, and completion continuity. The principal unresolved technical risk is the validity of selected cure-to-blast dependencies, which require geotechnical confirmation before any removal or relaxation.

PART III - COMPETENCY AND REFLECTION

12. What Changed in the Planning Approach

Starting assumption	Final understanding
A busy resource is the bottleneck	A constraint must be proven through schedule impact and sensitivity.
Priority should force a task to start	Priority only ranks eligible tasks competing for the same resource.
Month-end inventory means resource backlog	Inventory may be normal lead time if it converts in the next reporting period.
More production equipment should increase tonnes	It cannot help when stopes are not physically ready or are predecessor-blocked.
A task date before the data date means it is complete	Actual status must be entered; planned dates do not become actual automatically.
Finish date is the only schedule KPI	Monthly tonnes, readiness, coverage, milestones and continuity are equally important.
Custom event quantity can count a milestone	Native Finish In / Start In task counts are more reliable for event inventory.
Deswik.Sched is mainly STP	It supports detailed STP and MTP scheduling, but not strategic optimization alone.

13. Competency Gained

Competency area	Evidence from the training
Schedule architecture	Built WBS and executable task packages for 25 stopes, 3 panels, 4 horizons and 5 levels.
Dependency design	Created access, readiness, production, fill and cure relationships and identified unverified geotechnical links.
Resource modeling	Configured composite resource pools, assignment rates, calendars and active versus elapsed work.
Resource leveling	Used descending priorities, cleared and re-levelled scenarios, and interpreted task eligibility.
Data quality	Detected zero-priority packages through report evidence and traced their production impact.
Production reporting	Built monthly development, drilling, bogging and paste-fill profiles with validated totals.
Readiness analytics	Created current-state inventories from cumulative Finish In and Start In counts.
Bottleneck analysis	Confirmed development capacity through one-variable sensitivity and rejected false constraints.
Forecast control	Updated progress, re-levelled from the data date, and separated milestone variance from completion variance.
Management communication	Prepared final KPI, assumptions, risks and technical conclusions.

14. Errors and Troubleshooting That Produced the Most Learning

Issue	What it taught
Read-only CSV fields	Use import for network structure, then configure natively.
Zero driving quantity with assigned resource	Active tasks require a valid driving quantity.
Incorrect assignment rate	Always inspect effective assignment-level rate and task duration.
Combined paste and cure task	Separate active resource work from elapsed waiting.
Priority = 0 on active packages	Create a standing audit filter for unintended zero priorities.

Issue	What it taught
Progressive custom event count	Use Task Count (Finish In/Start In) for discrete gates.
Unrelevelled schedule after a change	Re-level before interpreting dates, queues or bottlenecks.
Temporary readiness inventory	Check conversion next month before declaring a resource constraint.

15. Limitations and Remaining Learning

- All quantities, rates, resource quantities and durations are synthetic training assumptions.
- The model was not linked to validated Deswik.CAD geometry because only SViz was available.
- Cure-to-blast relationships require geotechnical review and physical adjacency/stress evidence.
- Ventilation, traffic interaction, shaft/orepass capacity, haulage congestion and maintenance were simplified.
- Cost, NPV, workforce rosters, contractor availability and budget constraints were not optimized.
- Stage 13 recovery planning was designed but not completed as an independent controlled-delay experiment.
- A future training should separate MTP from detailed weekly STP execution and from strategic LTP optimization.

15.1 Recommended next training

Next module	Purpose
Underground weekly STP control	Shift/weekly production targets, actuals, exceptions, equipment availability and execution reforecast.
Geotechnical sequence validation	Translate adjacency, fill strength and exposure rules into justified predecessors.
CAD-Schedule integration	Link task packages to mine geometry and validate physical location logic.
Cost and budget integration	Add unit costs, crew availability, contracting and monthly budget constraints.
Open-pit LTP workflow	Pit phases, pushbacks, mining/processing balance, fleet strategy and strategic scheduling tools.

APPENDICES

Appendix A - Final Schedule Audit Checklist

1. Schedule is re-levelled after the latest change.
2. No active resource-consuming task has unintended Priority = 0.
3. Every resource assignment uses the correct effective rate table.
4. Active tasks have a non-zero driving quantity; cure and milestones do not consume resources.
5. Development Ready precedes Drill Ready, then Production Drilling, Blast, Bogging, Paste Fill, Cure and Complete.
6. Parent and summary tasks are excluded from physical quantity totals.
7. Full-model report totals equal 6,225 DEV m, 625 slot m, 70,929 drill m, 1,986 Mt and 674,491 m3 fill.
8. Readiness inventories are non-negative and cumulative gate counts end at 25.
9. Baseline is protected; all tests and updates are performed in separate files.
10. Unverified geotechnical predecessors remain flagged for engineering confirmation.

Appendix B - Quick Reference: Report Inventory Logic

Question	Use
How many tasks entered a state this month?	Task Count (Finish In) on the upstream gate
How many tasks left the waiting state?	Task Count (Start In) on the downstream activity
How many are waiting only?	Cumulative upstream Finish - cumulative downstream Start
How many are waiting or currently processing?	Cumulative upstream Finish - cumulative downstream Finish
How many are currently processing?	Cumulative downstream Start - cumulative downstream Finish
How much broken ore remains?	Cumulative blasted tonnes - cumulative progressive bogged tonnes
How much prepared coverage exists?	Prepared tonnes / average planned bogging of the next 3 months

Appendix C - Key Deswik.Sched Operating Rules

- Clear leveling before comparing a changed scenario with its baseline.
- High priority does not override predecessors, calendars, or resource eligibility.
- Use one-variable sensitivity to prove a resource constraint.
- Do not increase equipment before correcting rates, priorities, calendars and dependencies.

- Do not use parent roll-ups together with child production quantities in the same total.
- Use an elapsed calendar for passive curing and working calendars for active work.
- Use the data date to protect history and level only remaining future work.
- Interpret a report field according to its time-cycle behavior, not only its name.

Appendix D - Final Evidence Files

File / evidence	Purpose
UG_MTP_Stage14_Management_Report	Final management summary and conclusions
UG_MTP_Stage14_Report_Pack	Monthly and readiness reports, executive dashboard, audit and assumptions
Monthly Reports Baseline.csv	Baseline monthly physical production profile
Monthly Reports Updated Forecast.csv	Progress-updated and re-levelled monthly profile
Readiness Reports Baseline.csv	Baseline cumulative gates and current inventories
Readiness Reports Updated Forecast.csv	Updated readiness profile
UG_MTP_25_Stope task and predecessor files	Original synthetic network and schedule structure
Synthetic mine layout images and DXF package	Spatial learning aids; not validated operating design

Closing Statement

The complete training progressed from basic task entry to a defensible underground MTP schedule. The strongest outcome was not the final date alone. It was the ability to explain why the schedule behaves as it does, identify whether a problem is data, sequence, lead time, or resource capacity, and support every conclusion with reports and controlled tests.